Xi'an Jiaotong-Liverpool University

西交利物浦大学

MODULE CODE	EXAMINER	DEPARTMENT	TEL
CPT101		COMPUTER SCIENCE	

1st SEMESTER 2022/23 RESIT EXAMINATIONS

BACHELOR DEGREE – Year 2

COMPUTER SYSTEMS

TIME ALLOWED: 2 Hours

INSTRUCTIONS TO CANDIDATES

1. **This is an open-book exam.**
2. **Answer all questions.**
3. **Answers should be written in the answer booklet(s) provided.**
4. **No calculator is allowed during the examination.**
5. **Only answers in English are accepted.**

THIS PAPER MUST NOT BE REMOVED FROM THE EXAM HALL.

28. (20 marks) Supply correct arguments and instructions to the missing places **(i)** to **(x)** in the following program segment that ensures the user enter a number greater than 10 otherwise the program requests the user to input again.

```
int number;
(i) format[] = "%d";
_asm{
(ii):
  (iii) eax, number ; load number's memory address
  push eax          ; push onto stack
  lea   ebx, (iv)
  push  (v)         ; push onto stack
  call  (vi)        ; read user input
  add   esp, (vii)    ; return stack pointer
  mov   eax, (viii)   ; move user input into eax
  cmp   eax, (ix)     ; compare with set condition
  (x) readNumber  ; jump if condition not met
}
```

END OF PAPER